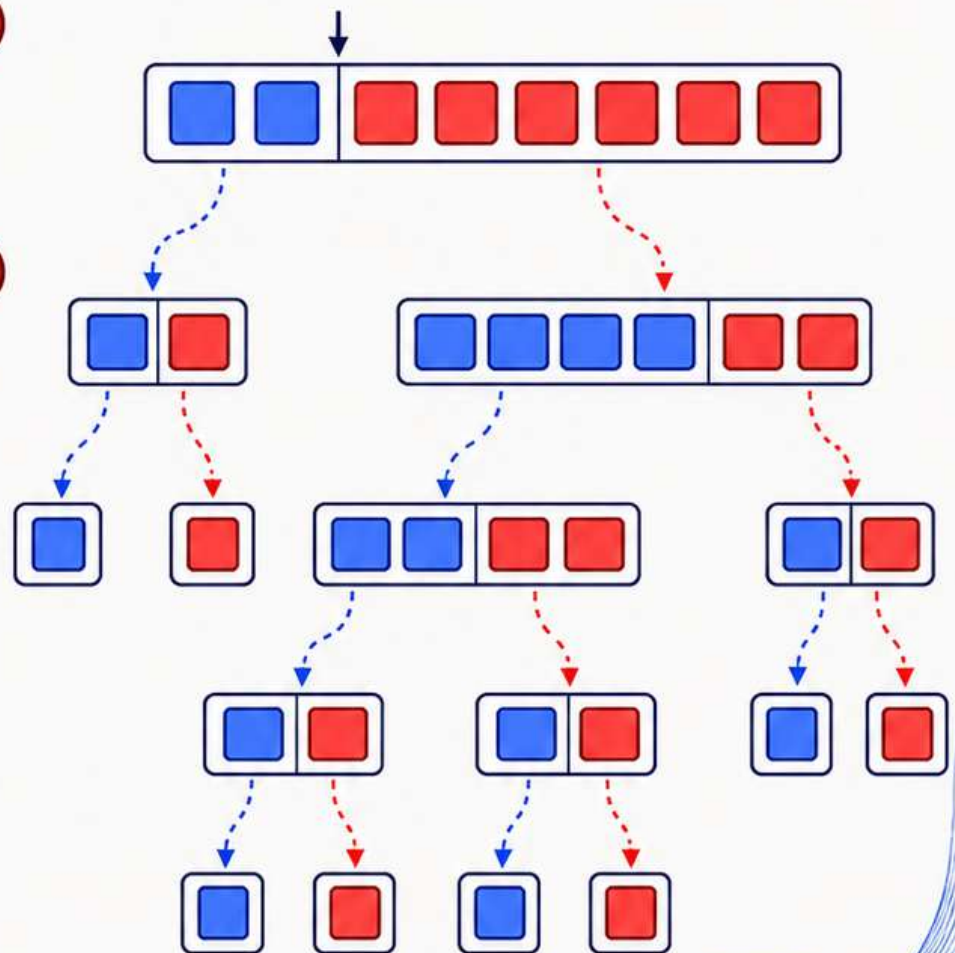
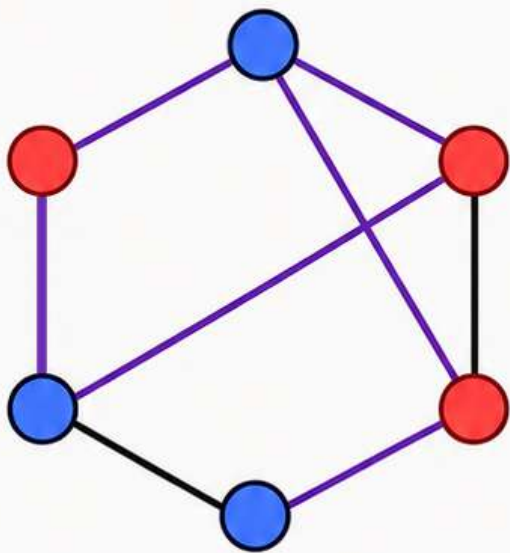


Randomised Algorithms

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Appendices

Appendix A

Probability and Expectation

A.1 Linearity of Expectation

For any random variables $X_1, X_2, \dots, X_n$ (not necessarily independent),

$$\mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i]$$

A.2 Tail Sum Formula

For a non-negative integer-valued random variable X ,

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \Pr(X \geq k) = \sum_{k=0}^{\infty} \Pr(X > k)$$

For a non-negative continuous random variable X ,

$$\mathbb{E}[X] = \int_0^{\infty} \Pr(X \geq t) dt$$

A.3 Union Bound

For any events $A_1, A_2, \dots, A_n$,

$$\Pr\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \Pr(A_i)$$

This is often used to upper bound the probability that at least one “bad event” occurs.

A.4 Independence of Expectation

If X and Y are independent random variables, then

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

A.5 Variance

For a random variable X , define the variance of X as

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

For any two random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) \quad (\text{where } \text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y])$$

If X and Y are independent, then $\text{Cov}(X, Y) = 0$, so

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

More generally, for any collection $\{X_1, \dots, X_n\}$,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j)$$

A.6 Indicator Variables

For an event A , define the indicator random variable

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

$$\implies \mathbb{E}[I_A] = \Pr(A)$$

A.7 Law of Total Expectation

Let X and Y be random variables. Then

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$$

provided the expectations exist. Intuitively, the theorem states that we may first compute the expected value of X under every possible circumstance described by Y , and then average those conditional expectations according to the probability of each Y .

For a series of random variables $X_0, X_1, \dots$,

$$\mathbb{E}[X_{i+1}] = \mathbb{E}[\mathbb{E}[X_{i+1} \mid X_i]]$$

A.8 Wald's Equation

Definitions

Let $X_1, X_2, \dots, X_N$ be i.i.d. random variables with finite expected value μ .

$$\mathbb{E}[X_1] = \mathbb{E}[X_2] = \dots = \mathbb{E}[X_N] = \mu$$

Now let the number of X 's in the series (which is of course, N) also be a random variable with expected value n .

Valid Stopping Time

For N to be a valid stopping time for a series of random variables $X_1, X_2, \dots$ the decision of whether to stop at i (implying $N = i$) must only depend on the information gathered until X_i . This is fairly intuitive since the decision to stop can not be based off future information.

For example, a rule like " $N = i$ if $X_i < \mu$ " is valid but the rule " $N = i$ if $X_{i+1} > \mu$ " is not.

The Equation

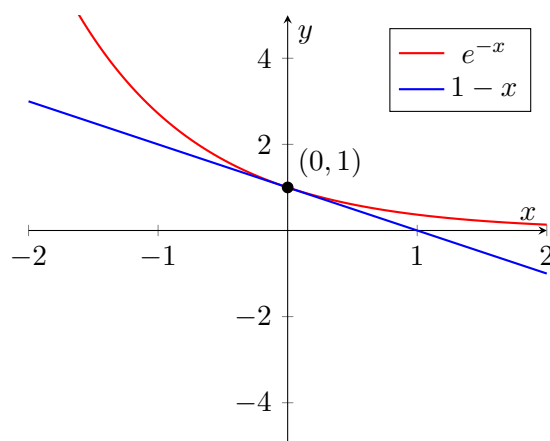
Given that N is a valid stopping time for the series of random variables $X_1, X_2, \dots$ the Wald's Equation states that the expected sum equals the expected number of terms multiplied by the expected value of each term.

$$\mathbb{E} \left[\sum_{i=1}^N X_i \right] = \mathbb{E}[N] \cdot \mathbb{E}[X] = n \cdot \mu$$

Appendix B

Inequalities

B.1 Exponential Inequality



$$1 - x \leq e^{-x} \quad \forall x \in \mathbb{R} \implies (1 - x)^k \leq e^{-kx} \quad \forall k \in \mathbb{N}, x \in \mathbb{R}$$

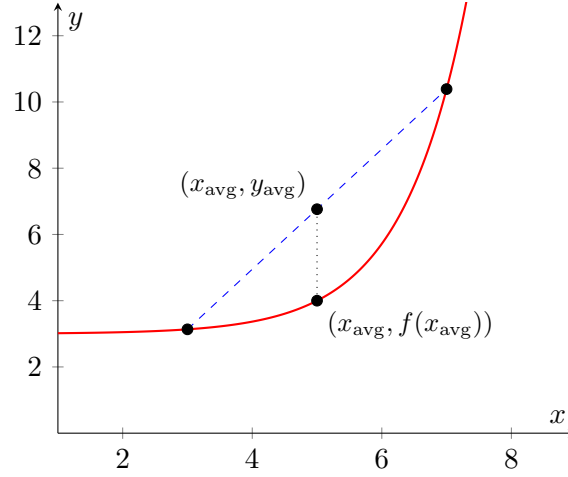
B.2 Jensen's Inequality

For a concave-up function f and random variable X ,

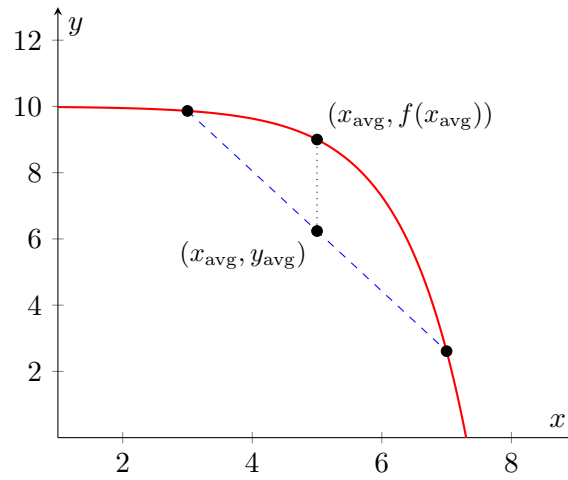
$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

Another form of Jensen's inequality is for a convex function f and real numbers $x_1, x_2, \dots, x_n$,

$$f\left(\frac{1}{n} \sum_{i=1}^n x_i\right) \leq \frac{1}{n} \sum_{i=1}^n f(x_i) \iff f(x_{avg}) \leq y_{avg}$$



For concave-down functions, the inequality is reversed, i.e. $f(x_{avg}) \geq y_{avg}$.



B.3 Markov's Inequality

Let X be a non-negative random variable and let $a > 0$. Then

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof

Define the indicator variable I as

$$I = \begin{cases} 1 & \text{if } X \geq a, \\ 0 & \text{otherwise.} \end{cases}$$

Then $X \geq aI$ follows by definition! Taking expectations,

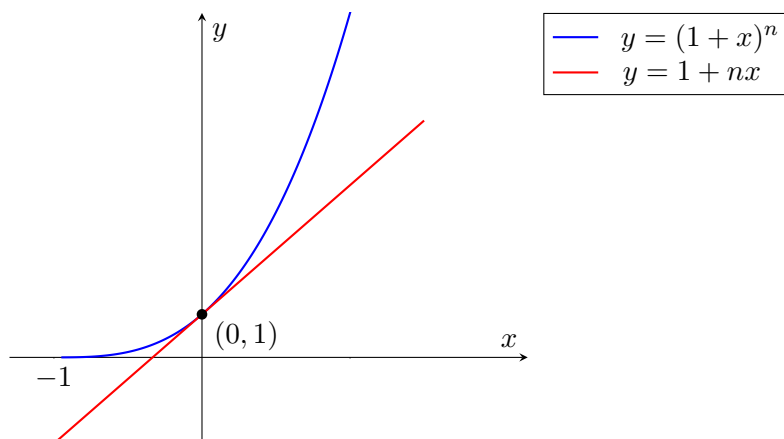
$$\mathbb{E}[X] \geq a \cdot \mathbb{E}[I] = a \cdot \Pr(X \geq a) \implies \Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

An equivalent and useful form is $\Pr(X \geq c \mathbb{E}[X]) \leq \frac{1}{c}$

B.4 Bernoulli's Inequality

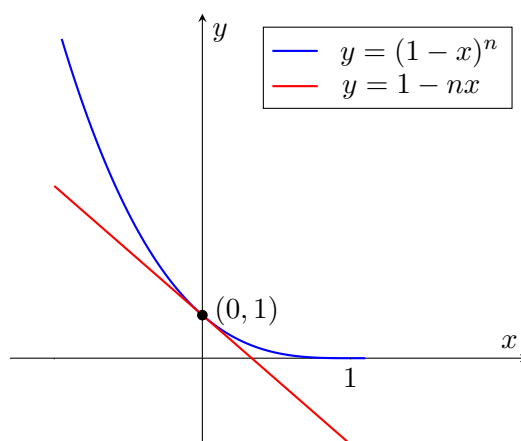
Let $x \geq -1$ be a real number and $n \geq 1$ be an integer. Then $(1+x)^n \geq 1+nx$.

$$x \geq -1 \text{ and } n \geq 1 \implies (1+x)^n \geq 1+nx$$



A commonly used equivalent form is obtained by substituting $x \rightarrow -x$.

$$x \leq 1 \text{ and } n \geq 1 \implies (1-x)^n \geq 1-nx$$



Appendix C

Convergence and Limit Theorems

It is quite difficult to properly grasp what these theorems say if you do not know the precise definitions of the terms used to state them. That is why I decided to be more elaborate in this appendix because there a lot of common misconceptions which I struggled to get clarity on until I took a rigorous and exhaustive approach.

C.1 Identical and Independently Distributed Random Variables

This is also denoted by i.i.d. for short. If it is stated that $X_1, X_2, \dots$ are i.i.d, then they are independent and have the same probability distribution. They are essentially copies of each other, but the value of one does not affect the others.

C.2 Almost Surely

When we say an event occurs “almost surely” we mean that it is true with probability one. The phrase “almost surely” can be used within a sentence like "X is almost surely equal to Y" which means "X is equal to Y" almost surely. We can also say that an event being almost sure is equivalent to the event having “almost every” outcome in its set.

This does not necessarily mean that every outcome is part of the event, it just means the "measure" of the set of outcomes in the event is equal to the total measure of the sample space. We will not be rigorously defining measure here but let me demonstrate via an example.

A random real number chosen from $[0, 1]$ is almost surely not $\frac{1}{2}$. This is because the measure of the set $[0, 1] - \{\frac{1}{2}\}$ is equal to the measure of the sample space.

C.3 Almost Never

When we say an event occurs “almost never” we mean that it is true with probability zero. Again, this does not mean that the event has no outcomes in its set, but rather that the measure of that set is zero as compared to the measure of the sample space. The complementary event of an almost never event is an almost sure event.

The example we used earlier can be restated as "a random real number chosen from $[0, 1]$ is almost never $\frac{1}{2}$ ".

C.4 $\lim_{n \rightarrow \infty} X_n = Y$

The event $\lim_{n \rightarrow \infty} X_n = Y$ is defined (where all X_n and Y are random variables) as the set of all outcomes ω such that $\lim_{n \rightarrow \infty} X_n(\omega) = Y(\omega)$. Since the sequence $X_n(\omega)$ is a real number sequence, the usual definition of limit applies.

In case Y is a constant a then it is the set of all outcomes ω such that $\lim_{n \rightarrow \infty} X_n(\omega) = a$. I personally do not like this notation since we are trying to define the notion of convergence and hence limits for random variables and in the definition we are using the notation $\lim_{n \rightarrow \infty} X_n$ which makes it seem circular when it really is not. The notation means to talk about an event defined using the limit of sequences of real numbers and not a sequence of random variables.

C.5 Convergence in Probability

A sequence of random variables X_n "converges in probability" to a constant a if for every $\varepsilon > 0$, the probability of X_n deviating from a by more than ε tends to zero.

$$X_n \xrightarrow{P} a \iff \lim_{n \rightarrow \infty} \Pr(|X_n - a| > \varepsilon) = 0 \quad \forall \varepsilon > 0$$

Note that we can **not** in general, swap the limit and the probability.

C.6 The Weak Law of Large Numbers

Let $X_1, X_2, \dots, X_n$ be i.i.d random variables with $\mathbb{E}[X_i] = \mu$ and finite variance σ^2 . Define the random variable Z_n to be average of the random variables $X_1, X_2, \dots, X_n$.

$$Z_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

The Weak Law of Large Numbers (WLLN) states that Z_n converges in probability to μ .

$$n \rightarrow \infty \implies Z_n \xrightarrow{P} \mu$$

Proof

For independent random variables, the variance of their sum is the sum of the variances.

$$\text{Var}(X_1 + \dots + X_n) = n\sigma^2$$

Dividing by n^2 yields the variance of the average –

$$\text{Var}(Z_n) = \frac{\sigma^2}{n}$$

Applying Chebyshev's inequality (only valid if the variance is finite) gives us the probability of a deviation of the average from its expected value –

$$\Pr(|\bar{Z}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \implies \lim_{n \rightarrow \infty} \Pr(|Z_n - \mu| > \varepsilon) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\varepsilon^2} = 0$$

So the probability that Z_n deviates from μ by more than ε tends to zero for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \Pr(|Z_n - \mu| > \varepsilon) = 0$$

This is equivalent to Z_n converging in probability to μ , hence the Weak Law of Large Numbers is proved.

C.7 Almost Sure Convergence

A sequence of random variables X_n "almost surely converges" to a constant a if the event $\lim_{n \rightarrow \infty} X_n = a$ occurs almost surely. This means for almost every outcome ω , $\lim_{n \rightarrow \infty} X_n(\omega) = a$.

$$X_n \xrightarrow{\text{a.s.}} a \iff \Pr\left(\lim_{n \rightarrow \infty} X_n = a\right) = 1$$

Example

It is not yet intuitively obvious what is the difference between almost sure convergence and convergence in probability. Consider the sequence of independent random variables $X_1, X_2, \dots$ such that

$$X_n = \begin{cases} 1 & \text{with probability } \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

Let us consider the convergence in probability of X_n to 0. For $\varepsilon \geq 1$, $\Pr(|X_n - 0| > \varepsilon) = 0$ since X_n is at most 1 thus $|X_n - 0| \leq 1$. For $\varepsilon < 1$, $\Pr(|X_n - 0| > \varepsilon) = \frac{1}{n}$. In both cases, as $n \rightarrow \infty$, $\Pr(|X_n - 0| > \varepsilon) \rightarrow 0$. Hence X_n converges in probability to 0.

Now consider the almost sure convergence to 0 – the probability that the event $(X_n \neq 0)$ occurs is $\frac{1}{n}$ for all n . Applying the second Borel–Cantelli lemma on the event $(X_n \neq 0)$ we get $\Pr(X_n \neq 0 \text{ i.o.}) = 1$.

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty \implies \Pr(X_n \neq 0 \text{ i.o.}) = 1$$

Thus the sequence $X_n(\omega)$ almost surely has infinitely many n where $X_n(\omega) \neq 0$. From the concept of limits of real number sequences, a sequence of 0's and 1's has infinitely many 1's iff it can not converge to 0. Since it almost surely has infinitely many 1's, it almost never converges to 0.

So this properly disproves the almost sure convergence of the sequence X_n to 0. In fact, it is almost never convergent to 0, yet it is convergent in probability to 0. Hence the two notions are not the same.

With some thought, it can be shown that almost sure convergence implies convergence in probability, but as we saw, not necessarily the other way around.

C.8 A_n infinitely often

Let $A_1, A_2, A_3, \dots$ be a sequence of events. Then the event " A_n infinitely often" or " A_n i.o." is defined as there are infinite $n \in 1, 2, 3, \dots$ such that A_n occurs. Thus the event " A_n i.o." occurs iff for all N , at least one of the events $A_N, A_{N+1}, A_{N+2}, \dots$ occur.

$$A_n \text{ i.o.} = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$$

C.9 The Borel–Cantelli Lemmas

The **first** Borel–Cantelli lemma states that if the sum of $\Pr(A_n)$ is finite, then the event $(A_n \text{ i.o.})$ occurs almost never.

$$\sum_{n=1}^{\infty} \Pr(A_n) < \infty \implies \Pr(A_n \text{ i.o.}) = 0$$

The **second** Borel–Cantelli lemma states that if all A_n are independent and the sum of $\Pr(A_n)$ diverges, then the event $(A_n \text{ i.o.})$ occurs almost surely.

$$\text{All } A_n \text{ independent and } \sum_{n=1}^{\infty} \Pr(A_n) = \infty \implies \Pr(A_n \text{ i.o.}) = 1$$

C.10 The Strong Law of Large Numbers

Let $X_1, X_2, \dots, X_n$ be i.i.d random variables with $\mathbb{E}[X_i] = \mu$ and finite variance σ^2 . Define the random variable Z_n to be average of the random variables $X_1, X_2, \dots, X_n$.

$$Z_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

The Strong Law of Large Numbers (SLLN) states that Z_n converges almost surely to μ .

$$X_n \xrightarrow{\text{a.s.}} \mu$$

The proof of the SLLN uses the Borel–Cantelli lemmas and we will not state it here as the proof itself does not hold much instructive value.

Application of SLLN

If I want to find the expected value of a random variable X and the only thing I can do is sample the variable $X \dots$ then I can approximate the expected value really well by averaging over all the samples. This is because sampling a random variable X is equivalent to creating a random variable X_1 and then evaluating it at a random outcome ω . Sampling it n times is equivalent to creating n i.i.d. random variables $X_1, X_2, \dots, X_n$ and then evaluating them at a random outcome ω .

Taking the average on the n samples would mean evaluating the average random variable Z_n at ω –

$$Z_n(\omega) = \frac{X_1(\omega) + X_2(\omega) + \dots + X_n(\omega)}{n}$$

Assuming n really is very large, we can use the Strong Law of Large Numbers to state that Z_n converges almost surely to the expected value of X_1 . This is equivalent to saying for almost every outcome ω , $Z_n(\omega)$ converges to $\mathbb{E}[X_1]$. In practice for well-behaved random variables this does work quite well if the variance is not too large. It is usually used in physics for experimental data.

If we want more guarantees about the distribution of the average Z_n , we can use the Central Limit Theorem.

C.11 Convergence in Distribution

Let X_n be a sequence of random variables with distribution functions $F_n(x) = \Pr(X_n \leq x)$. We say that X_n converges in distribution to a random variable X , written as

$$X_n \xrightarrow{d} X,$$

if the distribution functions satisfy

$$F_n(x) \rightarrow F(x)$$

This is the weakest form of convergence. Convergence in probability implies convergence in distribution.

C.12 The Central Limit Theorem

If $X_1, X_2, \dots$ are i.i.d. with expected value μ and finite variance σ^2 and Z_n be their average, then

$$\frac{\sqrt{n}}{\sigma} (Z_n - \mu) \xrightarrow{d} \mathcal{N}(0, 1)$$

This means that the distribution of the "normalized" average approaches the standard normal distribution as $n \rightarrow \infty$, even though the average itself approaches μ . Here the normalisation factor is $\frac{\sqrt{n}}{\sigma}$. Interestingly, the Central Limit Theorem is enough to prove the Weak Law of Large Numbers.

What can we conclude from the Central Limit Theorem? We can say that the average of a large number of samples have fluctuations from the mean of order $\frac{\sigma}{\sqrt{n}}$. In other words the error in the average is $\Theta(\frac{1}{\sqrt{n}})$. Since we know the approximate distribution of the average, we can find things like the value of the probability of it deviating from the expected value by some ε instead of using weak general upper bounds like Chebyshev.

Appendix D

Discrete Mathematics

D.1 Pigeonhole Principle

Let $a_1, a_2, \dots, a_n$ be non-negative real numbers and let

$$S = \sum_{i=1}^n a_i, \text{ then there exists an } i \text{ such that } a_i \geq \frac{S}{n}$$

This simply states that at least one value is at least the average value.

Proof by Contradiction

If every $a_i < \frac{S}{n}$, then summing over all i gives

$$\sum_{i=1}^n a_i < S$$

which is a contradiction.

D.2 Relations

A relation $R : A \rightarrow B$ is defined as the set of $(a, b) \in A \times B$. The presence of an element (a, b) in R is denoted by $a R b$.

D.2.1 Partial Orders

A relation $\preceq$ on a set S is called a *partial order* if it satisfies the following properties for all $a, b, c \in S$:

1. **Reflexivity:** $a \preceq a$
2. **Antisymmetry:** $a \preceq b$ and $b \preceq a \implies a = b$
3. **Transitivity:** $a \preceq b$ and $b \preceq c \implies a \preceq c$

Posets

A *poset*, is a pair $(S, \preceq)$ consisting of a set and a partial order on it.

D.2.2 Equivalence Relations

A relation $\equiv$ on a set S is called an *equivalence relation* if it satisfies the following properties for all $a, b, c \in S$:

1. **Reflexivity:** $a \equiv a$
2. **Symmetry:** $a \equiv b \implies b \equiv a$
3. **Transitivity:** $a \equiv b$ and $b \equiv c \implies a \equiv c$

A very common construction of an equivalence relation is from a partial order as follows –

$$aRb \text{ iff } a \preceq b \text{ and } b \preceq a$$

Equivalence Classes

Given an equivalence relation $\equiv$ on S , the *equivalence class* of an element $a \in S$ is all the elements related to it.

$$[a] = \{x \in S : x \equiv a\}$$

D.3 Graph Theory

A graph G is an ordered pair (V, E) where V is a set of vertices and $E \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is a set of edges. $|E|$ is usually denoted by m and $|V|$ is usually denoted by n .

D.3.1 Degree

For a vertex $v \in V$, the degree $d(v)$ is the number of edges which have v as an endpoint. Equivalently, $d(v)$ is the number of neighbours of v .

Given n is the number of vertices, the average degree of a graph is

$$d = \frac{1}{n} \left(\sum_{v \in V} d(v) \right) = \frac{2m}{n}$$

D.3.2 Bipartite Graph

A graph $G = (V, E)$ is bipartite if the vertex set V can be partitioned into two disjoint sets L and R such that

$$V = L \cup R, \quad L \cap R = \emptyset$$

and every edge connects a vertex in L to a vertex in R . That is, there are no edges with both endpoints in the same set.

D.3.3 Subgraph

A subgraph of a graph (V, E) is a pair of sets (V', E') where $V' \subseteq V$ and $E' \subseteq E$ with the additional condition that the subgraph must be a valid graph!

D.3.4 Clique

A clique in a graph G is a subset chosen from the set of vertices of G such that every pair of vertices in the clique is connected by an edge. The size of the clique is defined as the number of vertices in the clique.

Appendix E

Linear Algebra

E.1 Vector Spaces

Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set V equipped with vector addition and scalar multiplication satisfying the usual axioms:

- closure under addition and scalar multiplication,
- associativity and commutativity of addition,
- existence of a zero vector,
- existence of additive inverses,
- distributive properties,
- compatibility of scalar multiplication.

An example of a common vector space is the set of n -tuples of elements from $\mathbb{F}$,

$$\mathbb{F}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{F}\}$$

Common choices for the field $\mathbb{F}$ include the real numbers $\mathbb{R}$ or the integers $\mathbb{Z}$ with the usual addition and multiplication, and the finite field $\mathbb{F}_p$ where p is a prime with addition and multiplication modulo p .

E.2 Subspaces

A subset $W \subseteq V$ is called a subspace of the vector space V if

1. $0 \in W$
2. for all $u, v \in W$, we have $u + v \in W$
3. for all $u \in W$ and $\alpha \in \mathbb{F}$, we have $\alpha u \in W$

E.3 Dimension of a Subspace

A collection of vectors $v_1, v_2, \dots, v_k$ is called a basis of a subspace W if

- the vectors are linearly independent
- any vector in W is a linear combination of the vectors

The number of vectors in any basis of W is called the dimension of W and is denoted by $\dim(W)$.

E.4 The Nullspace

Let $M \in \mathbb{F}^{m \times n}$. The nullspace of M is defined by

$$\mathcal{N}(M) = \{x \in \mathbb{F}^n : Mx = 0\}.$$

The nullspace is also called the *kernel* of M , and we write

$$\ker(M) = \mathcal{N}(M).$$

The nullspace of a matrix is a subspace of $\mathbb{F}^n$.

E.5 Nullity and Rank

Since the nullspace is a subspace, it has a well-defined dimension.

Define $\text{nullity}(M) = \dim(\ker(M))$.

The rank of M , denoted $\text{rank}(M)$ is the dimension of the column space or row space of M , equivalently the maximum number of linearly independent columns or rows of M .

E.6 The Rank–Nullity Theorem

One of the fundamental results of linear algebra is the rank–nullity theorem.

Theorem (Rank–Nullity). For every matrix $M \in \mathbb{F}^{m \times n}$, we have

$$\text{rank}(M) + \text{nullity}(M) = n.$$

An immediate consequence is that if $M \neq 0$, then

$$\text{rank}(M) \geq 1, \implies \text{nullity}(M) \leq n - 1.$$